

Assignment 1: AI System Specification

Course: Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

Student Name: Neeraj Kumar

Roll No: 2K23/CSM/103

Submitted to: Sir Rajesh Kumar

Department: Computer Science, University of Sindh

Task 1: PEAS Specification

Application: A Smart Traffic Signal Controller for reducing congestion in cities.

Performance Measure:

1. Average wait time: Reduction in average seconds vehicles spend idling at red lights.
2. Throughput: Number of vehicles successfully passing through the intersection per hour.
3. Emergency response delay: Ensuring zero delay for approaching ambulances or fire trucks.
4. Fuel efficiency: Estimated reduction in vehicle emissions due to decreased idling.

Environment: The system operates at a busy urban intersection. Key external factors include erratic human driving behaviors, unpredictable pedestrian crossings, changing weather conditions (rain, snow) affecting road friction, and emergency vehicle occurrences.

Actuators:

1. Traffic light signals: Changing the state of Red, Yellow, and Green lights for all lanes.
2. Pedestrian crossing signals: Activating 'Walk' or 'Don't Walk' indicators.
3. Digital road signs: Displaying speed limits or congestion warnings to approaching drivers.

Sensors:

1. Inductive loop detectors: Embedded road sensors to detect vehicle presence and weight.
2. High-mounted optical cameras: Visual monitoring of lane queues and pedestrian activity.
3. Acoustic receivers: Detecting sirens or specialized radio signals from emergency vehicles.

Task 2: Environment Classification

Fully Observable vs. Partially Observable: Partially Observable. Justification: The sensors can only see vehicles currently near the intersection, not the traffic building up several blocks away that might arrive soon.

Single-agent vs. Multi-agent: Multi-agent. Justification: The traffic controller must coordinate its timing with other smart traffic controllers at adjacent intersections to create green wave corridors.

Deterministic vs. Stochastic: Stochastic. Justification: Human drivers are unpredictable; cars may stall, run red lights, or pedestrians may jaywalk randomly.

Episodic vs. Sequential: Sequential. Justification: Extending a green light now directly impacts the queue length and wait times of the cross-traffic in the next minute.

Static vs. Dynamic: Dynamic. Justification: Traffic flow, vehicle arrivals, and pedestrian movements are constantly changing while the AI is computing the next light cycle.

Discrete vs. Continuous: Continuous. Justification: Vehicle speeds, distances, and optimal light timing intervals exist on a continuous spectrum of physical time and space.

Task 3: Critical Analysis

1. Greatest Technical Challenge: The greatest technical challenge is the Stochastic and Dynamic nature of human behavior combined with the Multi-agent requirement. Unlike a controlled factory robot, a traffic AI must deal with irrational human actions (jaywalking, speeding, accidents). Furthermore, optimizing one intersection might accidentally cause a massive bottleneck at the next intersection down the road. Therefore, designing a reinforcement learning algorithm that successfully cooperates with neighboring intersection agents in real-time, while instantly adapting to sudden local anomalies (like a stalled car), is highly complex.

2. Utility Function and Trade-offs: Utility Function: $U = (w_1 * \text{Vehicle_Throughput}) - (w_2 * \text{Pedestrian_Wait_Time})$ This utility function handles the trade-off between maximizing vehicle flow and ensuring pedestrian convenience. If the weight of pedestrian wait time (w_2) was doubled, the agent's behavior would change to heavily favor foot traffic. It would cycle to red much faster the moment a pedestrian pushes the crosswalk button. While this makes the intersection extremely safe and fast for walkers, it would severely fragment vehicle flow, causing massive traffic jams for cars during rush hour.

Structured Representation

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        "justification": "Must coordinate with other traffic AIs at adjacent intersections."

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